Alessio Amaolo

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EDUCATION

Princeton University – Ph.D. Chemistry and Materials. Advised in Electrical and Computer Engineering. — Expected 2026/2027

Princeton University (3.9/4.0 GPA) – M.A. Chemistry (Pass with distinction) — May 2023

- Selected coursework: Mathematical Methods of Engineering, Numerical Algorithms for Scientific Computing (Python, C++), Nanophotonics (MPB/Scheme), Quantum Optics, Condensed Matter Physics, Quantum Field Theory

California Institute of Technology (3.9/4.0 GPA) — BSc Chemistry June 2021

- Selected graduate coursework: Computational Physics (Python), Quantum Computation, Advanced Condensed Matter Physics, Advanced Inorganic Chemistry
- Selected undergraduate coursework: Computing Systems (C, Assembly), Programming Methods (C++), Bayesian Statistics (R)

PUBLICATIONS – Google Scholar

10. **Amaolo, A.,** Maldonado, T.J., Chao., P., Strekha, B., Mohajan, J., Chen, F., Molesky, S., Rodriguez, A.W. Optimal inverse design in nanophotonics: a tutorial. *Advances in Optics and Photonics (Invited review paper).* In preparation.

9. Virally, P., Molesky, S, Chao, P., **Amaolo, A.,** Rodriguez, A.W., Bounds on ordered singular values of the Maxwell Green's function. In preparation.

8. Molesky, S., Chao, P., **Amaolo, A.**, Rodriguez, A.W. Inferring structure via duality for photonic inverse design. *arXiv:2504.14083.* 2025.

7. Chao, P.*, **Amaolo, A.*,** Molesky, S., Rodriguez, A.W. Bounds as blueprints: towards optimal and accelerated photonic inverse design. *arXiv:2504.10469.* 2025. Submitted to Optica.

6. **Amaolo, A.*,** Chao, P.*, Strekha, B., Clarke, S., Mohajan, J., Molesky, S., Rodriguez, A.W. Maximum Shannon capacity of photonic structures. *arXiv:2409.02089.* 2024. Submitted to Optica.

5. Strekha, B., **Amaolo, A.,** Mohajan, J., Chao, P., Molesky, S., Rodriguez, A.W. Limitations on bandwidth-integrated passive cloaking. *Physical Review A.* 2024.

4. **Amaolo, A.,** Chao, P., Maldonado, T.J., Molesky, S., Rodriguez, A.W. Physical limits on Raman scattering: the critical role of pump and signal co-design. *Physical Review A Letters.* 2024.

3. Maldonado, T.J., Pham, D.N., **Amaolo, A.,** Rodriguez, A.W., Türeci, H.E. Negative electrohydrostatic pressure between superconducting bodies. *Physical Review B.* 2024.

2. **Amaolo, A.,** Chao, P., Maldonado, T.J., Molesky, S., Rodriguez, A.W. Can photonic heterostructures provably outperform single-material geometries? *Nanophotonics. 2024.*

1. McNicholas, B.J., Nie, C., Jose, A., Oyala, P.H., Takase, M.K., Henling, L.M., Barth, A.T., **Amaolo, A.,** Hadt, R.G., Solomon, E.I., Winkler, J.R., Gray, H.B., Despagnet-Ayoub, E. Boronated Cyanometallates. *Inorg. Chem.* 2022.

RESEARCH EXPERIENCE

Student Researcher, Google Deepmind — Summer 2025

Research on computational lithography and mask process correction.

Graduate Researcher in Rodriguez Group, Princeton — Present

Performing research at the intersection of convex optimization and photonics focusing on calculating fundamental limits on photonic device performance.

- Developing novel algorithms for photonic inverse design utilizing convex relaxations, currently beating state-of-the-art algorithms 10x in performance in 10x fewer iterations.
- Formulated first information-theoretic treatment of photonic devices to calculate optimal electromagnetic information transfer, including in antennas and waveguides (e.g., fiber-optics).
- Extended methods of calculating performance limits for photonic multi-material, non-linear optical, and cloaking devices.
- Developed a differentiable, GPU-compatible FDFD Maxwell solver in JAX for large-scale photonic optimization.

Undergraduate Research Fellow, Chan Group, Caltech — Summer 2020

Performed research on tensor networks for simulating many-body quantum physics.

- Worked on algorithmic optimizations for evaluating the energies of highly interacting systems.
- Implemented various novel tensor network contraction techniques in Python.

Undergraduate Research Fellow, Gray Group, Caltech — Summer 2019

Researched the ligand field theory and electronic structure of boronated cyanometallates via spectroscopic methods.

Undergraduate Research Fellow, Gray Group, Caltech — Summer 2018

Researched methods of electrochemical reduction of carbon dioxide to produce oxygen.

TECHNOLOGIES

Proficient in Python (+NumPy, SciPy), scientific computing and ML in JAX, C/C++, MPI, OpenMP.

AWARDS

Jane Street Graduate Research Fellowship Finalist — 2025

- Finalist, invited to present at poster session

Princeton Chemistry Department David V. Milligan '62 Fellowship — 2021-2022

- Awarded by faculty in recognition of academic excellence

Reed and Ruth Brantley Research Fellowship — Summer 2020

Doris S. Perpall SURF Speaking Competition semifinalist (top 10% of presentations) — 2019

Donald Voet and Jerome Vinograd Research Fellowship — Summer 2019

Dr. and Mrs. Daniel C. Harris Research Fellowship — Summer 2018

TEACHING

Teaching Assistant, **Optical and Quantum Electronics** – Held help sessions and wrote questions for a graduate course on light-matter interactions for applications in electronics. Teaching reviews. — Fall 2024

Teaching Assistant, **Electromagnetism for Engineers** – Held two weekly labs focusing on applications of electromagnetism to engineering. Teaching reviews. — Spring 2024, Spring 2025

Teaching Assistant, **Ecology** – Held three weekly labs focusing on experimental techniques in ecology. Prepared midterm and final exam questions. Teaching reviews. — Fall 2023

Teaching Assistant, **Electromagnetism** – Held regular problem-solving sessions, office hours, and labs for a first-year advanced course on electromagnetism. Teaching reviews. — Spring 2023

Teaching Assistant, **General Chemistry** – Held three weekly recitations with a mix of lecturing and problem solving. Prepared midterm and final exam questions. Teaching reviews. — Fall 2022

Teaching Assistant, **Statistical Mechanics** – Held weekly recitations, office hours, and an exam review session for an undergraduate course in statistical mechanics. Teaching reviews. — Spring 2021

Teaching Assistant, **Physical Chemistry** – Held weekly recitations and graded problem sets for a course on quantum chemistry. Prepared notes for the second part of the course. Fall 2019

Selected Student Teaching Feedback

- *Electromagnetism for Engineers:* "Alessio was extremely helpful, cared about our understanding of the material, and was a very fair and kind individual."
- *Statistical Mechanics:* "Alessio was an amazing TA. He was always very kind in the way he answered questions without making me feel like it was a dumb question. One of the best physics TAs I've encountered."
- *General Chemistry:* "I feel as if most of my understanding of the course came from Alessio and his inherent ability to teach a group of students effectively."